9. Playfair Cipher — Decrypt the PT-109 Message

CODE:

```
def create_matrix(key):
    alphabet = "ABCDEFGHIKLMNOPQRSTUVWXYZ"
    letters = ""

    for ch in (key.upper() + alphabet):
        if ch.isalpha():
            ch = "I" if ch == "J" else ch
            if ch not in letters:
                letters += ch

    return [list(letters[i:i+5]) for i in range(0, 25, 5)]


def decrypt_playfair(ciphertext, matrix):
    pos = {matrix[r][c]: (r, c)
           for r in range(5) for c in range(5)}

    result = ""

    for i in range(0, len(ciphertext), 2):
        a, b = ciphertext[i], ciphertext[i+1]
        r1, c1 = pos[a]
        r2, c2 = pos[b]

        if r1 == r2:
            result += matrix[r1][(c1-1) % 5]
            result += matrix[r2][(c2-1) % 5]
        elif c1 == c2:
            result += matrix[(r1-1) % 5][c1]
            result += matrix[(r2-1) % 5][c2]
        else:
            result += matrix[r1][c2]
            result += matrix[r2][c1]

    return result


key = input("Enter Playfair key: ")

ciphertext = (
    "KXJEYUREBEZWEHEWRYTUHEYFSKREHE"
    "GOYFIWTTTUOLKSYCAJPOBOTEIZONTX"
    "BYBNTGONEYCUZWRGDSONSXBOUYWRHE"
    "BAAHYUSEDQ"
)

ciphertext = ciphertext.upper().replace("J", "I")
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if len(ciphertext) % 2 != 0:
    ciphertext += "X"

matrix = create_matrix(key)
print("Decrypted message:", decrypt_playfair(ciphertext, matrix))
```

OUTPUT:

